

# Nikhil Patnaik

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## Profile Summary

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Engineering undergraduate specializing in Data Science with a strong foundation in software development, API design, and data-driven systems. Experienced in building full-stack applications using Python, Java, SQL, and modern web frameworks, with hands-on exposure to back-end architecture, database design, and role-based access systems. Has explored Artificial Intelligence concepts including deep learning, computer vision, NLP, and Generative AI, with practical experience in prompt engineering and LLM integration. Strong problem-solving ability, comfortable working in fast-paced environments with evolving requirements, and quick to translate ideas into working technical solutions. Continuously learning emerging technologies to build scalable, real-world systems.

## Education

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### B.Tech in Computer Science Engineering (specialization in Data Science)

Sikkim Manipal Institute of Technology, Majitar, Sikkim

2023 – 2027

CGPA (as of 6th semester) : 7.25/10

### Class 12 (Higher Secondary)

Odisha Adarsh Vidyalaya, Nabarangpur, Odisha

2020 – 2022

Percentage : 81.2%

### Class 10 (Secondary)

DAV Public School

2019 – 2020

Percentage : 78%

## Professional Experience

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### Software Engineering Intern

Accops Systems Pvt. Ltd. (May 2026 to July 2026)

- Designed and built 20+ REST APIs (FastAPI) covering authentication, RBAC, content management, and reporting
- Architected PostgreSQL schema with Alembic migrations; built sequential workflow engine with quiz logic, scoring, and resume-state management
- Translated evolving business requirements into technical solutions through direct stakeholder collaboration
- Independently deployed and managed the app on a Linux VM, including environment setup and troubleshooting
- Integrated a locally-hosted LLM (Ollama/Llama3) to build an AI assistant chatbot for employee queries
- Followed Git-based feature branch workflow with PR reviews; leveraged AI tools to accelerate development

## Technical Skills

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- **Programming:** Python, Java, C
- **Backend & APIs:** FastAPI, REST API Design, SQLAlchemy, Alembic, JWT Authentication
- **Databases:** SQL, PostgreSQL, MongoDB
- **Web Development:** HTML, CSS, JavaScript, React.js, Node.js, Express.js
- **Artificial Intelligence & Machine Learning:** Scikit-learn, PyTorch, TensorFlow/Keras, CNN basics, Prompt Engineering, LLM Integration (Ollama)
- **NLP & Generative AI:** TF-IDF, NLTK, Hugging Face Transformers, RAG fundamentals

- **Data Science & Analysis:** Pandas, NumPy, Data Preprocessing, Feature Engineering, Exploratory Data Analysis
- **Tools & Technologies:** Git, VS Code, Postman, Linux server administration

## Projects

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- **Content-Based Movie Recommendation System**

*Tech Stack: Python, Pandas, NumPy, Scikit-learn, TF-IDF, Cosine Similarity, NLTK*

*Dataset: tmdb\_5000\_movies.csv, tmdb\_5000\_credits.csv*

- Preprocessed and merged movie metadata to build a unified dataset
- Applied TF-IDF vectorization and cosine similarity for recommendation generation
- Handled cold-start scenarios using metadata-based filtering

- **Lung Cancer Detection using CNN**

*Tech Stack: Python, TensorFlow/Keras, OpenCV, NumPy*

*Dataset: LUNA-16k (subsets)*

- Built a CNN model for classification of lung nodules from CT scan images
- Performed image preprocessing and feature scaling to improve model input quality

- **Signature Forgery Detection using GANs**

*Tech Stack: Python, TensorFlow/PyTorch, OpenCV, NumPy*

- Designed a GAN-based model to distinguish between genuine and forged signatures
- Applied adversarial training for feature learning and pattern generation

- **Sentiment Analysis on Restaurant Reviews**

*Tech Stack: Python, Pandas, NumPy, VADER Sentiment Analysis, Scikit-learn, Matplotlib, Seaborn*

*Dataset: Custom dataset of 200+ Google reviews from 21 restaurants in Majhitar-region [East Sikkim]*

- Applied VADER to classify 200+ reviews into sentiment categories
- Built a dish-based recommendation system using keyword filtering and sentiment scoring

## Soft Skills

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- Project planning and coordination
- Strong presentation and communication skills
- Calm under pressure and able to keep team members focused
- Problem-solving mindset
- Patience and understanding

## Certifications & Experiences

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- Summer Training on **Full Stack Web Development (MERN)** from E&ICT Academy, IIT Kanpur
- **GenAI: Prompt Engineering & AI Agents Micro-Certification** by PwC Advisory Launchpad Program

## Extracurricular Activities

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- Secured **3rd Position** in **Smart India Hackathon (SIH) 2024 Prelims**
- General Secretary of **Student Research Forum (SRF), SMIT**;  
Published 2 research blogs on Medium via SRF.
- Alumni Representative - **Student Council (Dept. of AI & DS)**